Successive h-indices using OpenAlex

1 Methodology

1.1 Data source and filtering

The underlying data is OpenAlex's public authors snapshot (Priem, Piwowar, & Orr, 2022), distributed as Parquet on `s3://openalex/data/parquet/authors/` (`us-east-1`, no credentials required) and partitioned into over 1,600 daily `updated_date=` directories totalling about 53 GB. Only four columns are read from each remote file (`id`, `h_index`, `affiliations`, `topics`), and an author is retained only if all of the following hold:

- `summary_stats.h_index` is non-null and greater than zero;
- at least one entry in `affiliations` has `institution.type = 'education'`;
- at least one entry in `topics` carries a non-null field classification.

Applying these filters to the full snapshot yields 17.1 million authors, spanning 20,669 institutions and OpenAlex's current list of 26 top-level fields.

1.2 Author-level assignment

Each retained author is reduced to a single row: one institution and one field. An author may list several affiliations with `institution.type = 'education'`; rather than fractionally splitting or duplicating that author across institutions, we assign them to the one with which they were most recently affiliated. Recency is read from OpenAlex's `affiliations` field, which records, for each institution an author has ever listed, the calendar years in which they published while there; we take the institution whose most recent such year is latest, breaking ties (identical latest year) by the numerically smallest OpenAlex institution identifier. We deliberately do not use the similarly-named `last_known_institutions` field for this: despite its name, it is simply the set of affiliations listed on an author's single most recent work, carrying no per-entry recency of its own, so when an author has more than one such affiliation (e.g., a joint appointment or co-affiliated coauthors misattributed to one author) there is no way to tell which is "more recent" than another.

Field assignment works the same way but is weighted. OpenAlex tags each author with a set of topics, each carrying a `count` (roughly, the number of the author's works classified under that topic) and a parent field. We sum `count` within each field and assign the author to whichever field carries the largest total weight: their modal field by publication volume, not simply their single most granular topic. The result is an `authors` table with one row per author: (`author_id, h_index, institution_id, institution_name, field, field_name`).

1.3 The h-index and its successive generalization

Hirsch's (2005) original index, here h_1, is computed by OpenAlex per author from citation counts and taken as given. Schubert's generalization (2007) lifts the same definition up a level of aggregation: a group of N members, each already carrying an h_{k-1} index, has an index $h_k = n$ if n of its members have $h_{k-1} \geq n$ and the remaining $N - n$ members have $h_{k-1} < n$:

Algorithm 1 Successive h-index of a group G, given a value $v(m)$ for each member $m \in G$

1: Sort G by v descending; let $r(m)$ be m's 1-indexed rank in that order
2: **return** $\max\{\, r(m) : v(m) \geq r(m),\ m \in G \,\}$

We use this rank-based procedure, implemented as a window function in SQL ranking rows within each group and taking the largest rank whose value has not yet fallen below it, for every successive index in this paper: h_2 of an institution over the h_1 of its authors, and h_3 of a country over the h_2 of its institutions.

1.4 h_2: institution and institution×field

h_2 is computed twice, over different groupings of the same author-level data. The field-specific table, `h2_by_institution_field`, groups authors by (institution, field) and yields 288,300 pairs; institutions are matched on their numeric identifier only; institution and field *names* are resolved afterward via `arg_max`, so that spelling or formatting variants of the same name across snapshot partitions cannot silently split one institution into several groups. The institution-overall table, `h2_by_institution`, ignores field entirely and applies the same procedure across all of an institution's authors regardless of discipline, yielding one h_2 per institution (20,669 rows).

1.5 h_3: country level

Institutions are mapped to countries using the `country_code` field of OpenAlex's institutions snapshot (also read directly from S3, joined on institution ID). h_3 is then computed with the same rank-based procedure, once globally per country over the h_2 values of its institutions (`h3_by_country`), and once per (country, field) over the field-specific h_2 values (`h3_by_field/`), producing one file per field.

1.6 Supplementary metrics

Several secondary statistics are derived from the h_2 and h_3 tables to probe what the headline numbers obscure:

- **Gini coefficient**, computed on the within-field distribution of institutional h_2, quantifies whether a field's excellence is concentrated in one institution (Gini near 1) or broadly distributed (Gini near 0).
- **Efficiency** normalizes for scale: $\mathrm{h}_2/\sqrt{\text{author\_count}}$ at the institution level, and $\mathrm{h}_3/\log_2(\text{institution\_count})$ at the country level. Both exponents follow from the self-consistency condition that defines a successive h-index: a group's index is the threshold x^* at which the number of members exceeding x^* equals x^* itself. If member values are drawn from a population with a power-law (Zipfian) tail $P(\text{value} > x) \sim x^{-a}$, this condition solves to $x^* \sim N^{1/(1+a)}$; taking $a = 1$, the classic exponent for individual bibliometric indices, predicts $x^* \sim \sqrt{N}$ for h_2 over authors. h_3 aggregates institution-level h_2 values, which are themselves already-smoothed aggregates with a thinner, closer-to-exponential tail; the same self-consistency argument with an exponential tail instead predicts $x^* \sim \log N$. Both predictions are then checked empirically by regressing the observed index against the predicted form: h_2 against $\sqrt{\text{author\_count}}$ ($r^2 = 0.82$) and h_3 against $\log_2(\text{institution\_count})$ ($r^2 = 0.75$).
- **Breadth score** counts, per institution, how many of the 26 fields it ranks in the global top 10, top 50, and top 100.
- **Non-biomedical ranking** recomputes institution-level h_2 after excluding authors whose modal field is Medicine, Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, or Health Professions, to test whether the overall ranking is, in substance, a ranking of medical school size.
- **Academic Olympics** converts each field's country-level h_3 ranking into an Olympic-style medal table: the top three distinct h_3 values in a field are gold/silver/bronze, ties at the same h_3 share a medal and the next tier is skipped (as in the real Olympics), and medals are tallied across all 26 fields per country.

1.7 Manual cross-validation

Because errors in OpenAlex's underlying author disambiguation or citation counts would propagate directly into every successive index, we manually spot-check a targeted, non-random sample of authors rather than a random one, chosen at the points where an error would be both most likely and most consequential: authors sitting exactly at an institution's h_2 threshold, where a ± 1 error in a single h_1 could change the institution's rank; top authors in fields where OpenAlex's topic classification is known to be thinner (e.g., Arts and Humanities); top authors at institutions with non-Anglophone naming conventions, where author-name disambiguation is more error-prone, including institutions suspected of affiliation-swapping for grant purposes (Bhattacharjee, 2011); and the small number of authors with the highest h_1 globally, as a sanity check at the extreme of the distribution. For each candidate we resolve the author's OpenAlex profile through the public API and compare it by hand against Web of Science, Scopus, and Google Scholar.

2 Results

2.1 Manual cross-validation

Of the 19 authors sampled under the scheme described in Subsection 1.7 (manual cross-validation), Table 1 summarizes agreement between OpenAlex's `h_index` and the corresponding value we located by hand in Web of Science, Scopus, and Google Scholar. Coverage itself varies sharply: Scopus returned a value for 16 of the 19 authors, Google Scholar for 12, and Web of Science for only 11.

Table 1 Agreement between OpenAlex and each external source, over authors for which that source returned a value. Diff is OpenAlex minus the external value.

Source	n	Mean diff	Mean abs. diff	Worst case
Web of Science	11/19	+27.2	29.0	Lei Jiang (+100)
Scopus	16/19	+6.7	13.9	Donald Small (+46)
Google Scholar	12/19	−21.9	34.6	A. Jafari (−85)

Scopus tracks OpenAlex most closely, both in coverage and in magnitude of disagreement. Google Scholar is systematically higher than OpenAlex, consistent with its broader crawl of preprints, theses, and other grey literature and its comparative permissiveness toward self-citation; this ordering (Google Scholar > Scopus > OpenAlex > Web of Science) reproduces cleanly among the five global top-h_1 sanity-check authors, none of whom show a discrepancy exceeding 27% of their OpenAlex value in either direction.

Four authors, however, show disagreement well outside this range. We read this as a signature of author-disambiguation failure (multiple distinct people merged under one OpenAlex author ID) rather than of genuine cross-database disagreement:

- **Donald Small** (Johns Hopkins): $h_1^{\text{OpenAlex}} = 103$ against Scopus = 57 and Google Scholar = 59.
- **Karen S. Calhoun** (University of Michigan): $h_1^{\text{OpenAlex}} = 45$ against Scopus = 11 and Google Scholar = 13.
- **Lei Jiang** (University of Science and Technology of China): $h_1^{\text{OpenAlex}} = 215$ against Web of Science = 115.
- **A. Jafari** (Sharif University of Technology): $h_1^{\text{OpenAlex}} = 139$, but Web of Science, Scopus, and Google Scholar report 41, 133, and 224 respectively.

All four fall in the non-Anglophone-institution or common-name strata of the sample, matching the concern raised in Subsection 1.7 that disambiguation is more error-prone for exactly these authors. Because such merges can only inflate an author's apparent h_1, and h_1 feeds directly into h_2 and h_3, a handful of merged identities at the right institution could measurably distort an institution- or country-level ranking; we flag this as an open limitation rather than one this sample can size definitively.

The two Arts and Humanities threshold authors sampled (Gillian Ramchand and G. Herbert Fowler, both sitting at Oxford's field-specific $h_2 = 30$ threshold) returned no value in any of the three external sources. This is consistent with the well-documented weaker coverage of book-based humanities scholarship in all major citation databases, OpenAlex included, and suggests that h-index comparisons in these fields should be read with correspondingly less confidence than in the sciences.

References

Bhattacharjee, Y. (2011, December). Saudi universities offer cash in exchange for academic prestige. *Science Magazine*, *334*, 1344–1345,

Hirsch, J.E. (2005). An index to quantify an individual's scientific research output. *Proceedings of the National Academy of Sciences*, *102*(46), 16569-16572, https://doi.org/10.1073/pnas.0507655102 Retrieved from https://www.pnas.org/doi/abs/10.1073/pnas.0507655102 https://www.pnas.org/doi/pdf/10.1073/pnas.0507655102

Priem, J., Piwowar, H., Orr, R. (2022). *OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts.* Retrieved from https://arxiv.org/abs/2205.01833

Schubert, A. (2007). Successive h-indices. *Scientometrics*, *70*(1), 201–205,